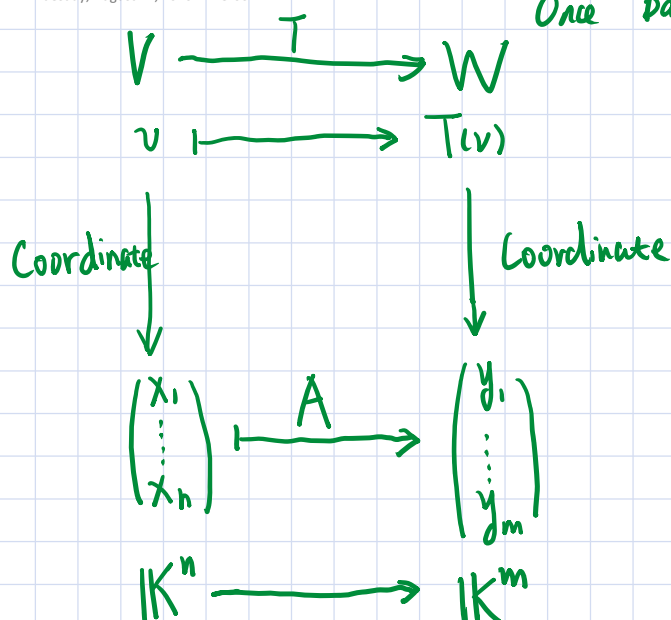


Once basis chosen:



- Basis Extension Lemma.

 $(V, +, \cdot)$ vec. space. and $v_1, \dots, v_k$ Lin. indep.Suppose $n = \dim V$ and $n > k$. ($n < \infty$)Then we can find $v_{k+1}, v_{k+2}, \dots, v_n \in V$ Lin. indep.and $v_1, \dots, v_k, v_{k+1}, \dots, v_n$ form a basis of V

Proof. is by an iterative argument.

Example: $\mathbb{R}^3$:

$$\begin{array}{c}
 \nearrow e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\
 \nearrow e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\
 \nearrow e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}
 \end{array}$$

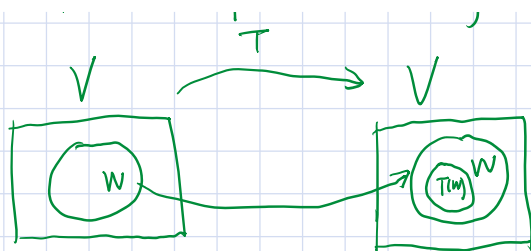
- Invariant Subspace of a linear transformation

 $T \in \mathcal{L}(V)$ ($T: V \rightarrow V$, linear)A subspace $W \leq V$ is invariant if

$$T(W) = \{T(w) : w \in W\} \subseteq W.$$

$$V \xrightarrow{T} V$$

Clearly V is an invariant



Clearly V is an invariant subspace for any $T \in \mathcal{L}(V)$ and $\{0_V\}$ as well.

- Two important invariant subspaces of $T \in \mathcal{L}(V)$
- ARE SUBSPACES
- $\text{Im } T = \{T(v) : v \in V\}$ ← What's the output of T
 - $\text{ker } T = \{v \in V : T(v) = 0_V\}$ ← What's being lost under T

Claim: For any $T \in \mathcal{L}(V)$, $\text{Im } T$ and $\text{ker } T$ are invariant.

Proof: Take $w \in \text{Im } T \subseteq V$, apply T , $T(w) \in \text{Im } T$ by defn.

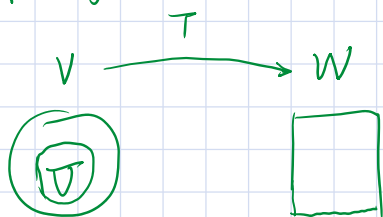
Take $u \in \text{ker } T$, then $T(u) = 0_V$.

Apply T : $T(T(u)) = T(0_V) = 0_V \Rightarrow T(u) \in \text{ker } T$.

$$T(0_V) = T(0_V + 0_V) = T(0_V) + T(0_V) \Rightarrow T(0_V) = 0_V$$

□

The above definition of $\text{Im } T$ and $\text{ker } T$ goes also for linear maps $\mathcal{L}(V, W)$ but the concept of invariant subspaces only works for $\mathcal{L}(V)$.



" $T(U) \subseteq U$ " doesn't make sense since $T(U) \subseteq W \neq V$.

In this case, $\text{Im } T \subseteq W$ and $\text{ker } T \subseteq V$.

- Rank, nullity and the rank-nullity Theorem.

- $\text{Rank}(T) := \dim \text{Im } T$ ← How many dimensions in W is the output of T ?

- $\text{nullity}(T) := \dim \text{ker } T$ ← How many dimensions in V is being mapped to 0_W by T ?

- Claim: $T \in \mathcal{L}(V, W)$ is injective $\Leftrightarrow \text{ker } T = \{0_V\}$
 $\Updownarrow$
 one-to-one

- Claim: (1) $T \in \mathcal{L}(V, W)$ is injective $\Leftrightarrow \ker T = \{0_V\}$
 $\Updownarrow$
 one-to-one
- (2) $T \in \mathcal{L}(V, W)$ is surjective $\Leftrightarrow \text{Im } T = W$
 $\Updownarrow$
 onto.

Proof: (2) is obvious.

(1) $\Rightarrow$: Suppose T injective and take $v \in \ker T$

$$T(v) = 0_W = T(0_V)$$

By injectivity, $v = 0_V$

$\Leftarrow$: Suppose $\ker T = \{0_V\}$.

If $T(u) = T(v)$, by linearity:

$$T(u-v) = 0_W \Rightarrow u-v \in \ker T = \{0_V\}$$

This means $u = v$.

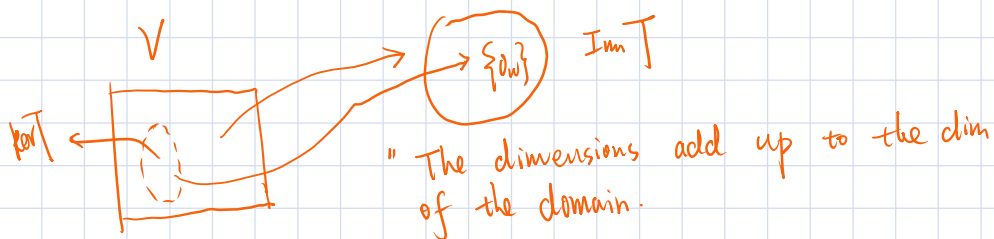
This proves T is injective.

?: How are $\text{Rank}(T) := \dim \text{Im } T$ and $\text{nullity}(T) := \dim \ker T$ related to each other? □

Ans: Rank-Nullity Theorem:

Let $T: V \rightarrow W$ linear. ($\dim V < \infty, \dim W < \infty$)

$$\text{Then } \dim V = \dim \text{Im } T + \dim \ker T.$$



$$\text{Equivalently: } \dim V = \text{Rank}(T) + \text{nullity}(T).$$

Proof: Let $u_1, \dots, u_k$ be a basis of $\ker T$ thus

$$\dim \ker T = \text{nullity}(T) = k.$$

$$u_1, \dots, u_k \in \ker T \subseteq V.$$

By Basis Extension we can find $v_1, \dots, v_r$, lin indep.

and $u_1, \dots, u_k, v_1, \dots, v_r$ is a basis for V .

$$\sim \quad 1, \dots, k \quad \quad 1, \dots, r.$$

and $\underline{u_1, \dots, u_k, v_1, \dots, v_r}$ is a basis for V .

$$\text{So } \dim V = k + r.$$

We want to show that $T(v_1), \dots, T(v_r)$ forms a basis for $\text{Im } T$.

First: $T(v_1), \dots, T(v_r)$ span $\text{Im } T$.

Take any $w \in \text{Im } T$, by defn. $\exists v \in V: w = T(v)$.

Since $\underline{u_1, \dots, u_k, v_1, \dots, v_r}$ is a basis for V

$$v = \underbrace{a_1 u_1 + \dots + a_k u_k}_{\in \ker T} + b_1 v_1 + \dots + b_r v_r \quad (*)$$

Apply T to both sides of $(*)$

$$\begin{aligned} w = T(v) &= T(a_1 u_1 + \dots + a_k u_k + b_1 v_1 + \dots + b_r v_r) \\ &= \underbrace{a_1 T(u_1) + \dots + a_k T(u_k)}_{0_w} + b_1 T(v_1) + \dots + b_r T(v_r) \\ &= \sum_{j=1}^r b_j T(v_j) \end{aligned}$$

$$\text{Thus } \text{Im } T \subseteq \text{Span}(T(v_1), \dots, T(v_r))$$

$\supseteq$: This is obvious since T is linear.

$$\text{Therefore } \text{Im } T = \text{Span}(T(v_1), \dots, T(v_r)).$$

Now, WTS. (= want to show) $T(v_1), \dots, T(v_r)$ lin. indep.

$$\text{Suppose } c_1 T(v_1) + \dots + c_r T(v_r) = 0_w$$

$$\text{By linearity } 0_w = T(c_1 v_1 + \dots + c_r v_r)$$

$$\text{This means } \underline{c_1 v_1 + \dots + c_r v_r} \in \ker T.$$

But recall that $u_1, \dots, u_k$ is a basis for $\ker T$

$$c_1 v_1 + \dots + c_r v_r = d_1 u_1 + \dots + d_k u_k.$$

$$\Rightarrow c_1 v_1 + \dots + c_r v_r - d_1 u_1 - \dots - d_k u_k = 0_v$$

Since $v_1, \dots, v_r, u_1, \dots, u_k$ is a basis $V \Rightarrow$ lin. indep.

$$\underline{c_1 = \dots = c_r = d_1 = \dots = d_k = 0_k}$$

This shows that $T(v_1), \dots, T(v_r)$ is lin. indep.

Lin. indep + spans $\text{Im } T \Rightarrow T(v_1), \dots, T(v_r)$ is basis for $\text{Im } T$

$$\text{Rank}(T) = \dim \text{Im} T = \# \text{ basis elements} = r.$$

$$\Rightarrow \dim V = k+r = \dim \ker T + \dim \text{Im} T \\ = \text{nullity}(T) + \text{Rank}(T). \quad \square$$

- Plan:
- ① Take basis for $\ker T$ in V
 - ② Extend it to a full basis of V
 - ③ Show that $\{T(\text{added basis elements})\}$ forms a basis of $\text{Im} T$
 - ④. Dimension counts follows.

Application:

Let $T \in \mathcal{L}(V, W)$ and $\dim V = \dim W$ ($\mathcal{L}(V)$ is a special case)

Then TFAE (the following are equivalent)

1) $\ker T = \{0_V\}$

2) T injective.

3) T surjective.

4) T bijective.

If $\ker T = \{0_V\}$, then by R-N:

$$\dim V = \dim \text{Im} T + \underbrace{\dim \ker T}_0 = \dim \text{Im} T$$

||

$\dim W$. This means $\text{Im} T = W$

because $\text{Im} T \leq W$

Then T is surjective

Surjective $\Rightarrow \dim W = \dim \text{Im} T$ plug into R-N.

$$\Rightarrow \dim \ker T = 0 \Rightarrow \ker T = \{0_V\}$$

- Rank of a matrix:

• Take $A \in K^{m \times n}$, then A naturally gives a linear map

$$T_A : K^n \longrightarrow K^m$$

$$x \longmapsto Ax$$

• What is Ax ? $\downarrow \quad \text{---} \quad \downarrow$

• What is Ax ?

$$Ax = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ a_{21}x_1 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{pmatrix}$$

$$= x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \dots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

Then $\text{Im } T_A = \{Ax : x \in K^n\}$

$$= \left\{ x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \dots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} : x_1, \dots, x_n \in K \right\}$$

$$= \text{Span} \left(\begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} \right)$$

$$:= \text{Col}(A)$$

Then we define.

$$\text{Rank}(A) = \dim \text{Im } T_A = \dim \text{Col}(A)$$

= # of independent Columns.

- Small note: $\dim \text{Col}(A)$ is technically speaking the column rank of A .

There is an analogous concept: row rank $\dim \text{Row}(A)$ which is the dimension of the Span of the rows of A

K_n

Fact: (Trust me bro!)

$$\dim \text{Col}(A) = \dim \text{Row}(A)$$

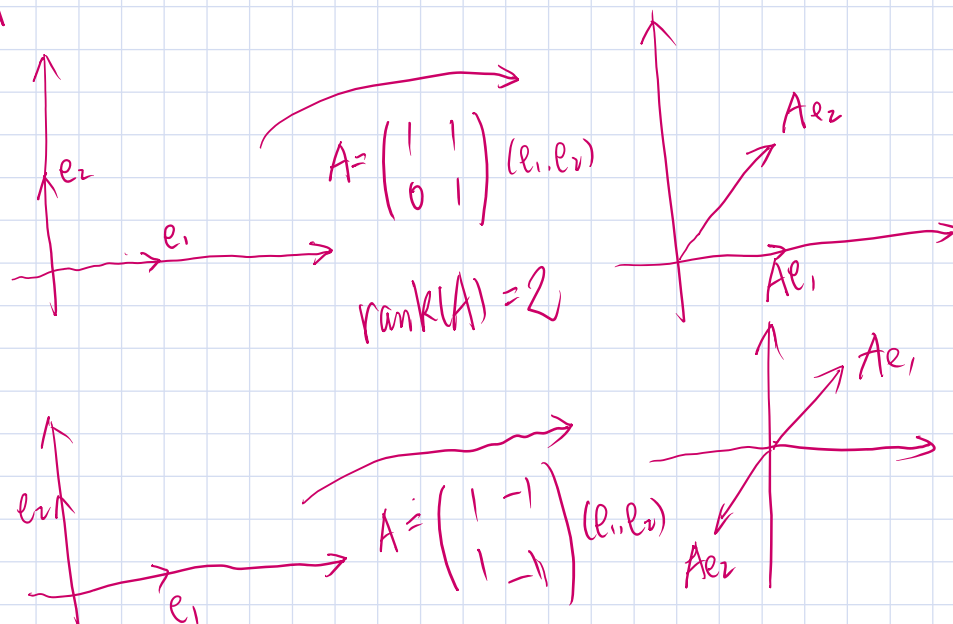
fact: $\dim \text{Col}(A) = \dim \text{Row}(A)$

$$\dim \text{Col}(A) = \dim \text{Row}(A)$$

Take: $\mathbb{R}^{2 \times 3} : \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \end{pmatrix}$ for example.

- Rank tells us how many lin indep. rows / columns there are in a matrix. Viewed as a linear map: how many indep. directions are being output.

Example: $\mathcal{L}(\mathbb{R}^2)$



- Change of Basis for a Linear map:

Domain: V Bases $\mathcal{E} = (e_1, \dots, e_n)$ and $\mathcal{E}' = (e'_1, \dots, e'_n)$

Codomain: W Bases $\mathcal{F} = (f_1, \dots, f_m)$ and $\mathcal{F}' = (f'_1, \dots, f'_m)$

Change of Basis: $(e'_1, \dots, e'_n) = (e_1, \dots, e_n) P$ (in V)
 $(f'_1, \dots, f'_m) = (f_1, \dots, f_m) Q$ (in W)

P invertible $n \times n$; Q invertible $m \times m$

P invertible $n \times n$; Q invertible $m \times m$

Take $v \in V$: Coord. vector $[v]_{\mathcal{E}} = P [v]_{\mathcal{E}'} \leftarrow$

$$\begin{aligned} \mapsto v &= (e_1, \dots, e_n) [v]_{\mathcal{E}} = (e_1, \dots, e_n) [v]_{\mathcal{E}'} \\ &= (e_1, \dots, e_n) P [v]_{\mathcal{E}'} \end{aligned}$$

Similarly $[w]_{\mathcal{F}} = Q [w]_{\mathcal{F}'}$ for any $w \in W$.

Let A be the matrix of T from $\mathcal{E} \rightarrow \mathcal{F}$

B " " " " from $\mathcal{E}' \rightarrow \mathcal{F}'$

Then: $B = Q^{-1} A P$.

To see this : for every $v \in V$.

$$[T(v)]_{\mathcal{F}} = A [v]_{\mathcal{E}} = A P [v]_{\mathcal{E}'} \quad (*) \leftarrow$$

On the other hand, do a change of basis for $T(v)$

$$[T(v)]_{\mathcal{F}} = Q [T(v)]_{\mathcal{F}'} = Q B [v]_{\mathcal{E}'} \quad (**) \leftarrow$$

Equate (*) and (**):

$$A P [v]_{\mathcal{E}'} = Q B [v]_{\mathcal{E}'}.$$

This holds for all vectors $[v]_{\mathcal{E}'}$ that is

all vectors in $\mathbb{K}^n$ then $AP = QB$

As Q is invertible : $B = Q^{-1} A P$.

$m \times n \quad m \times m \quad m \times n \quad n \times n$

As Q is invertible: $B = Q^{-1}AP$,
 $m \times n$ $m \times m$ $m \times n$ $n \times n$.

- When $B = Q^{-1}AP$ we call A and B are equivalent.
- Fact: Equivalent matrices have the same rank.
 And the rank is the rank of the linear map the matrix encodes.

Corollary: If $T \in \mathcal{L}(V)$ and $\mathcal{E} = (e_1, \dots, e_n)$, $\mathcal{E}' = (e'_1, \dots, e'_n)$,
 and $(e'_1, \dots, e'_n) = (e_1, \dots, e_n)P$

The matrix of T under $\mathcal{E}$ is A
 " " of T under $\mathcal{E}'$ is B

Then $B = P^{-1}AP$

$$\begin{aligned} [T(v)]_{\mathcal{E}} &= P[T(v)]_{\mathcal{E}'} \\ &= PB[v]_{\mathcal{E}'} \quad \parallel \end{aligned}$$

$$\begin{aligned} (e'_1, \dots, e'_n) &= (e_1, \dots, e_n)P \\ &\downarrow \text{coord.} \\ [v]_{\mathcal{E}} &= P[v]_{\mathcal{E}'} \end{aligned}$$

$$[T(v)]_{\mathcal{E}} = A[v]_{\mathcal{E}} = AP[v]_{\mathcal{E}'}$$

When $B = P^{-1}AP$ we say that A and B are similar. Note for $\mathcal{L}(V)$ the matrix is square: Only square matrices of the same size. can be tested whether they are similar.

• $B = Q^{-1}AP$ (Equivalence) for $m \times n$ A & B

- $B = Q^{-1}AP$ (Equivalence) for $m \times n$ $A \& B$
- $B = P^{-1}AP$ (Similarity) for $n \times n$ $A \& B$.

► Two matrices are equivalent iff they have the same rank.

→ under some given basis.
 ? : For a matrix A (describing linear map $T: V \rightarrow W$)

Can we find alternative bases for V and W .

such that $B = Q^{-1}AP$ is as simple as possible?

Ans: Yes! In fact:

$$\exists P \& Q \text{ invertible: } Q^{-1}AP = \begin{matrix} r & n-r \\ \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \end{matrix}_{m \times n}$$

$r = \text{Rank}(A) = \text{Rank}(T)$. Rank Normal Form

!! : For square matrices, A and B are similar iff _____.

"Ans": There is no simple answer.

We will see a case where for certain matrices $P^{-1}AP = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ (Diagonalization).

In general, the conditions are complicated and it depends on the field K .

Remark: If you are interested, search "Jordan Form" "Frobenius Form" of similar matrices.

- Sum and Direct Sum of Subspaces.

Take $V_1, \dots, V_K : V_i \leq V$ the sum:

$$V_1 + V_2 + \dots + V_K = \sum_{i=1}^K V_i = \{v_1 + v_2 + \dots + v_K : v_i \in V_i, \forall i=1, \dots, K\}$$

If $v \in \sum_{i=1}^K V_i$, then $v = v_1 + v_2 + \dots + v_K$ for some $v_i \in V_i, \forall i$.

If this representation is unique:

If $v \in \sum_{i=1}^K V_i$ then $\exists! v_1, \dots, v_K : v = v_1 + v_2 + \dots + v_K$

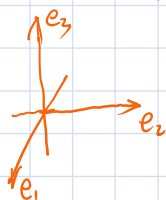
Then we call this sum a direct sum.

Denote it as $V_1 \oplus V_2 \oplus \dots \oplus V_K = \bigoplus_{i=1}^K V_i$.

Example: $\mathbb{R}^3 = \text{Span} \left\{ \begin{matrix} e_1 \\ e_2 \\ e_3 \end{matrix} \right\}$

For any $v \in \mathbb{R}^3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 e_1 + x_2 e_2 + x_3 e_3$

$\mathbb{R}^3 = \text{Span}\{e_1\} \oplus \text{Span}\{e_2\} \oplus \text{Span}\{e_3\}$.



But $\mathbb{R}^3 = \text{Span}\{e_1, e_2\} + \text{Span}\{e_2, e_3\}$
(Check this).

Claim: This sum is not direct.

Let's take $e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2} \left[\underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{\text{Span}\{e_1, e_2\}} + \underbrace{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}}_{\text{Span}\{e_2, e_3\}} \right] + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{2} \left[\underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}_{\text{Span}\{e_1, e_2\}} + \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{\text{Span}\{e_2, e_3\}} \right]$

So the rep. of e_2 is NOT UNIQUE.

So the sum is not direct by def.

- Why sum?

- Why sum?

Sum of vector spaces forms a vec. space.

(Check!)

Union of vec. spaces is usually NOT a vec. space.

Counterexample in $\mathbb{R}^2$: $\text{span}\{e_1\} \cup \text{span}\{e_2\}$.



Out of the union.

NOT A vec. space!

- Determinants.

! : Determinants only exists for square matrices.

What is the determinant?

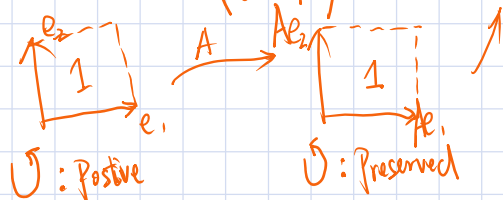
Take $T \in \mathcal{L}(\mathbb{R}^n)$.

$\det(T) = \text{signed volume } T([0,1]^n)$.

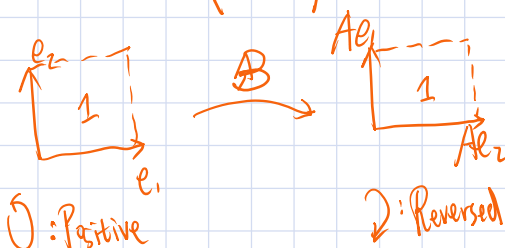
We've seen det. for 2×2 matrices.

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Consider: $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\det(A) = 1$



$$B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \det(B) = -1$$



$$C = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

$$\det(C) = 0$$

$$C = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}, \det(C) = 0$$



- What's generally true is that if A n.n.

$\Leftrightarrow A$ invertible.

→ $Ae_i = i^{\text{th}}$ column of A (Verify)

If curious / unclear. Watch 3 blue 1 brown on
Linear algebra.

- Change of basis for $T \in \mathcal{L}(V, W)$

= Sum and direct sum

Coming Next: Diagonalization & Inner product Spaces.